Introduction

- Many modern apps and speech-based systems collect and process users' voice data.
- This voice data often contains personal or sensitive information.
- Without proper protection, attackers can misuse it to steal identities or gain unauthorized access.

Domain Introduction

Audio Anonymization:
Hides or masks the speaker's identity while keeping the spoken content clear and understandable.

Audio Encryption:
Scrambles the entire audio so it becomes completely unreadable without decryption.

Use Case:
Enables safe sharing of speech data for daily life usage, research or AI conversion without revealing who spoke.

Literature Review

Paper 1

The VoicePrivacy 2022 Challenge: Progress & Perspectives in Voice Anonymisation

Authors: Pierre Champion, Nicholas Evans, Sarina Meyer, Xiaoxiao Miao, Michele Panariello, Massimiliano Todisco, Natalia Tomashenko, Emmanuel Vincent, Xin Wang, Junichi Yamagishi

Publication Date: 16 July, 2024

Publication: IEEE/ACM Transactions on Audio, Speech and Language Processing

Affiliation: EURECOM, France; Inria, France; University of Stuttgart, Germany

The VoicePrivacy 2022 Challenge: Progress & Perspectives in Voice Anonymisation

- This paper summarizes the VoicePrivacy 2022 Challenge, which evaluates techniques for anonymizing a speaker's identity in speech.
- It compares different anonymization approaches and shows a trade-off between privacy and speech quality.
- Achieving both strong privacy and high naturalness remains unsolved

The VoicePrivacy 2022 Challenge: Progress & Perspectives in Voice Anonymisation

- X-vector & ECPA to extract speaker identity
- Generate pseudo-speaker embedding with GAN
- Speech is resynthesized with TTS and HiFi-GAN
- Excellent privacy and good utility but loses pitch change requirement

Scope

- Human Vs Machine Understanding
- Dataset (multiple Speaker + bigger & more diverse dataset)
- Tradeoff between EER & WER

Paper 2

X-vector anonymization using autoencoders & adversarial training for preserving speech privacy

Authors: Juan M. Perero-Codosero, Fernando M. Espinoza-Cuadros, Luis A. Hernández-Gómez

Publication Date: January 2022

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Published in: Computer Speech & Language

X-vector anonymization using autoencoders & adversarial training for preserving speech privacy

Step 1: Feature Extraction

- **X-vector:** A digital "voiceprint" (embedding vector) that captures who you are.
- **F0 (Fundamental Frequency):** Your voice's pitch
- **Linguistic content:** What words/sounds you're saying

Step 2: Anonymize the X-vector → AAN (Autoencoder-Adversarial Network).

- **Autoencoder :** Tries to compress and reconstruct the x-vector
- **Adverserial Training :** Tries to erase identifying info like:
    a. Specific speaker identity
    b. Gender
    c. Accent

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Step 3: Generate New Anonymous Speech

- **The system uses:**
  - The anonymized x-vector (the identity)
  - Original pitch (F0)
  - Original Words (linguistic content)

To create new speech that says the same thing but sounds like a **different** person

X-vector anonymization using autoencoders & adversarial training for preserving speech privacy

Results

The paper compares its system to the "Baseline" system and other systems from the VoicePrivacy 2020 Challenge

- **Privacy (EER):** The paper's system achieved a very high EER (37.8%) in the main test scenario This was better (more private) than the Baseline (31.9%) and most other systems.
- **Intelligibility (WER):** The paper's system had a Word Error Rate (WER) of 11.0%. This was practically the same as the Baseline (10.9%) and the other systems.

X-vector anonymization using autoencoders & adversarial training for preserving speech privacy

Future Scope

More speaker characteristics for anonymization:

- Currently, they anonymize speaker identity, gender, and accent.
- In the future, they aim to include more characteristics like emotion, age, or speaking style.

Better anonymization of the full speech waveform:

- The current system anonymizes the x-vector (speaker embedding) but not the full speech waveform.
- In the future, they plan to apply the method directly to encoded speech and prosodic features (like rhythm and intonation) for deeper anonymization.

Paper 3

Quantized Approximate Signal Processing (QASP): Towards Homomorphic Encryption for audio

Authors: Tu Duyen Nguyen, Adrien Lesage, Clotilde Cantini, Rachid Riad

Publication Date: submitted on 15 May 2025, to

Computer Speech & Language

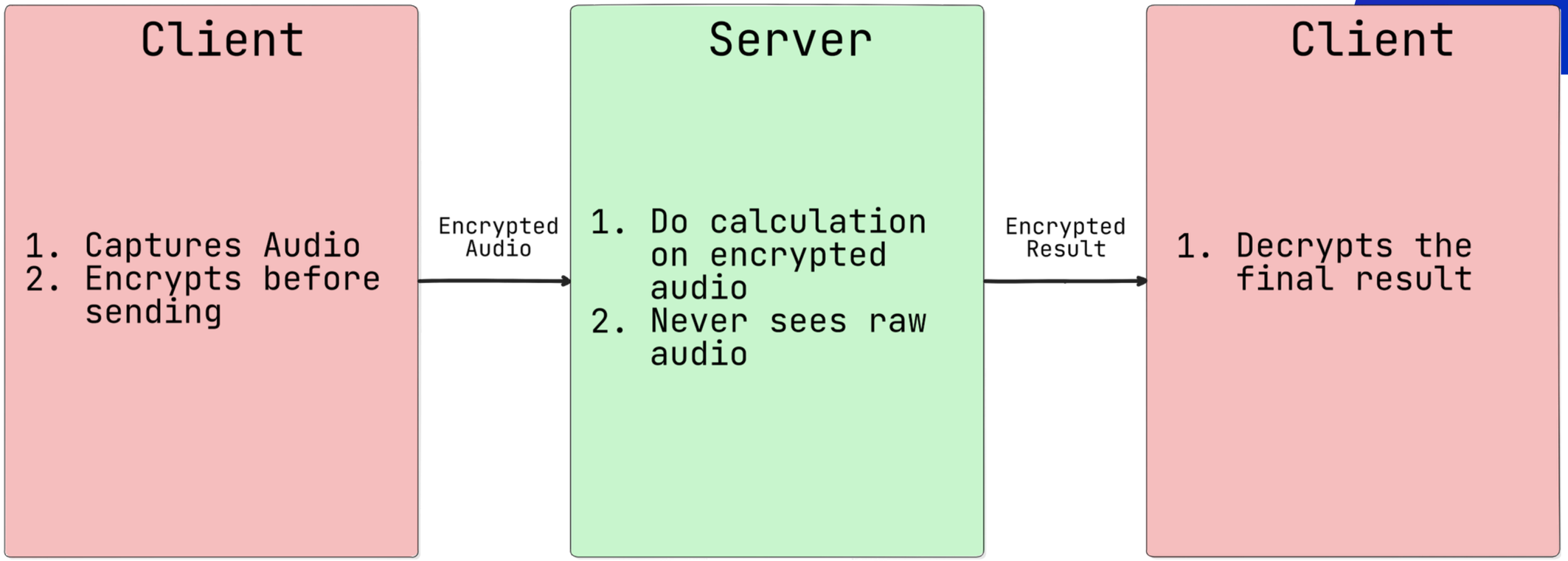
Affiliation: Callyope, a medical copilot AI company in Paris,

France.

Homomorphic Encryption (HE)

- A cryptographic method that allows computations on encrypted data.
- Can perform calculations on it without decryption.
- The server sees only encrypted file, but can still run functions (e.g., ML, statistics, MFCC, Mel-Spectogram)
- After computation, the result is decrypted by the owner and matches the output as if it was computed on original file.

Homomorphic Encryption (HE)



Limitations of Fully Homomorphic Encryption (FHE)

- Fully Homomorphic Encryption only allows computation on low-bit integers
- (usually <= 16 bits)
- Standard audio processing (STFT, Mel, MFCC) uses floating-point math and produces large intermediate values inexecutable inside HE.
- So direct manipulations (MFCC , Mels) on encrypted audio would cause overflow.
- It adds noise after every operation as can not calculate exact precision value and has higher computational cost.

QASP (Quantized Approximate Signal Processing)

- It basically converts the high presition continuous audio data to discrete low bit format.
- Does this By Quantization basically discretization of values so that higher calculations done easily
- Preserve only the necessary audio structure for downstream tasks.
- This discretization is achieved through agressive Quantization.

Future Scope

Possible Future Approach

- Anonymize the audio while keeping the privacy intact(tradeoff of EER & WER)
- Explore Homomorphic encryption technology
- Keep a key to reverse back to original version from anonymized one